

The Generation Half: Why Prompt-Routing Studies Understate Domain Specialization in Mixture-of-Experts Models

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Abstract

A widely cited reading of mixture-of-experts (MoE) language models holds that their experts do not specialize by subject matter: routing measured over supplied text spreads each expert across many topics, with chemistry, law, and the rest sharing the same few high-frequency experts. We show that this reading rests on half of the forward pass. A transformer routes tokens in two regimes: a parallel prefill pass that reads the prompt, and a step-by-step generation pass that writes the answer, and supplied-text routing is the prefill half. Across a suite of prompts spanning twenty academic domains, we identify the expert that receives the most routing weight in each domain, separately for each pass. Over prefill a single high-frequency expert wins almost every domain and routing looks domain-blind, reproducing the familiar negative picture. Over generation the winners disperse and most domains acquire their own leading expert. The shift is large and one-directional, survives a control that pads all prompts to a common length, and reproduces in a second, larger model whose expert indices are entirely different while the prefill-concentrated, generation-dispersed pattern holds. The upshot is a correction to how prior negative results should be read: prefill is the half the field measured, generation is the half it missed, and domain specialization in these models is largely a property of the generation pass.

Keywords: mixture-of-experts; expert specialization; routing; prefill; generation; autoregressive decoding; interpretability.

1 Introduction

Mixture-of-experts (MoE) layers replace a dense feed-forward block with many parallel expert blocks and a router that, for each token, selects a few of them. Only the selected experts run, so the model can hold a large parameter count while keeping per-token compute small. Because the router makes a discrete, inspectable choice at every token, it invites a natural question: do the experts carve knowledge up by subject, so that one becomes the chemistry expert and another the law expert? The most-cited look came up empty. Jiang et al. (2024), analyzing Mixtral’s routing on validation text, “do not observe obvious patterns in the assignment of experts based on the topic”;

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the structure they find is syntactic and positional. That negative result is load-bearing: it is widely read as evidence that MoE experts simply do not specialize by domain.

That look, and most like it, scored routing over supplied text. A transformer processes a sequence in two distinct regimes. During *prefill*, the whole prompt is processed in parallel to populate the key/value cache. During *generation*, output tokens are produced one at a time, each attending to everything before it. Supplied-text routing is the prefill regime. Routing happens in both passes, and the two are free to disagree. They disagree in a specific, consistent way, and that disagreement decides whether domain specialization is visible.

We route a domain-balanced prompt suite through two MoE models and, for each of twenty domains, identify the expert that receives the most routing weight, once over prefill positions and once over generated positions. Over prefill, one generalist expert wins 18 of 20 domains in the smaller model and 13 of 20 in the larger one, and routing looks domain-blind, reproducing the familiar picture. Over generation, the winners disperse: nearly every domain is won by its own distinct expert, and the generalist recedes. The same expert that looked like “the answer to everything” while the model read the prompt gives way to per-domain leaders once it begins to write.

What each expert encodes stays outside our scope, and we do not re-measure Mixtral. The result is about *where* in the forward process domain structure becomes measurable: prefill is the half the field measured, and generation is the half it missed. The canonical negative result holds for the prefill half it measured; it has been read too broadly, as a verdict on specialization, because supplied-text routing is domain-blind by construction of where it looks.

This paper makes three contributions. (1) We draw a methodological correction: reading expert specialization off prefill or supplied-text routing, or off routing pooled across passes, systematically understates it, which reframes how the field’s negative results on MoE specialization should be read. (2) We establish the effect that drives the correction: separating routing by forward-pass regime, prefill versus generation, places domain specialization in the generation pass and shows it largely absent from prefill, measured as the concentration of per-domain expert winners. (3) We show the effect survives a length-matched control that equalizes token counts across the two passes, and that it reproduces in a second, larger model whose expert indices do not correspond to the first.

2 Related work

Mixture-of-experts routing. The mixture-of-experts construction routes inputs to specialized subnetworks through a learned gate (Jacobs et al., 1991). Shazeer et al. (2017) made this sparse and scalable for deep networks with a top- k gate over thousands of experts, and the design was carried into large transformers by Switch Transformers (Fedus et al., 2022) and ST-MoE (Zoph et al., 2022). The models we study are open MoE checkpoints in the Qwen3 lineage (Yang et al., 2025). Across this line of work the router exposes an inspectable choice at every token, tempting a reading of it as a map of internal organization.

Do experts specialize by topic? This question is the paper’s starting point, and the most-cited answer is negative. Jiang et al. (2024) analyze Mixtral’s routing on validation text and report that they “do not observe obvious patterns in the assignment of experts based on the topic”: the expert distribution is similar across arXiv, biomedical, and philosophy documents, with structure instead in syntax and token position. The statement is carefully scoped to what it measured — routing over supplied text, which is the prefill regime — and it has since hardened, in looser uptake,

into the broader belief that MoE experts do not specialize by domain. Dai et al. (2024) reach a compatible intuition from the engineering side: they treat strong specialization as something to build in, through fine-grained expert segmentation and shared-expert isolation, which presumes that ordinary top- k routing specializes weakly. Our measurement extends the first result and qualifies the second. Read over prefill we, too, see little domain structure; the same probe read over the generation pass resolves into per-domain experts. The delta to Jiang et al. (2024) is the axis of measurement: the forward-pass regime.

Input- and token-dependent routing. Recent work shows routing depends strongly on the input. Avinash (2026) construct “routing signatures” and recover task identity from routing alone with high accuracy, establishing that the router is task-sensitive; and ichiro Hayashi et al. (2026), working with the same Qwen3.5-35B-A3B checkpoint we use, find that routing during code generation is strongly token-identity dependent and layer-dependent. Both condition on *what* is being processed (task or token), pooled within a regime; neither contrasts the prefill and generation passes for the same prompts, which is the comparison we make.

Prefill versus generation. The two-pass structure of transformer inference (a parallel prefill over the prompt followed by step-by-step decoding) is well established on the systems side, where the two phases have distinct compute and latency profiles and are increasingly served separately (Kwon et al., 2023; Zhong et al., 2024). We use the same distinction representationally: the two passes route to different experts, and the difference is where domain specialization becomes measurable.

3 Why prefill routing is a tempting place to look

There are practical reasons an analyst reaches for prefill routing first. Prefill is deterministic: the same prompt and the same weights produce the same router logits, with no sampling in the loop, so prefill statistics are exactly reproducible and easy to compare across runs. Prefill is also cheap to capture: a single forward pass over a short prompt yields one routing vector per token, no generation required. And prefill is where the prompt’s content actually sits, so it is intuitive to expect topical routing to be strongest exactly there, while the model is “looking at” the chemistry question.

The measurement below shows that this intuition is inverted. The content of the prompt is present during prefill, but the routing over prefill positions is dominated by a few generic experts and is largely blind to domain. The domain-specific structure appears later, while the model is producing domain-specific text. The reasons we can offer for this are partial (Section 6), but the empirical pattern is clean enough to state on its own and to act on.

4 Methods

Probe. The prompt suite contains 60 short questions covering 20 academic domains (archaeology, biology, chemistry, comparative religion, computer science, cybersecurity, economics, environmental science, history, law, linguistics, mathematics, medicine, neuroscience, philosophy, physics, political science, psychology, software engineering, statistics), with 3 questions per domain. Each question is a plain, domain-typical prompt (for example, a request to contrast two core concepts in the field).

Models. We use two MoE checkpoints from the same Qwen3.5 lineage (Yang et al., 2025), both 8-bit quantized fine-tunes (HauhouCS “uncensored” variants): a 35B-parameter model with $\approx 3\text{B}$ active per token, and a 122B-parameter model with $\approx 10\text{B}$ active. Both route each token to the top 8 of 256 experts per MoE layer. The two models have different layer counts and different internal layouts, so their expert indices are not comparable to each other; an expert numbered 114 in one model has nothing to do with expert 114 in the other. Decoding is greedy. Generation is capped (at 2,048–2,056 new tokens); we return to the consequences of that cap in Section 7.

Routing reconstruction. For each token we take the router’s pre-softmax logits over all 256 experts, apply softmax, keep the top 8, and renormalize over the selected experts so the eight weights sum to one. For an expert e on a fixed set of tokens (one pass of one prompt at one layer) we form three reductions of these renormalized weights: S_e , the fraction of tokens on which e is selected; Q_e , its mean weight on the tokens where it is selected; and W_e , its mean weight over all tokens (zero where unselected). Because all three reduce the same weights, $W_e = S_e Q_e$ holds exactly on any single token set; we use this only as a bookkeeping check that the selection mask, the conditional mean, and the marginal mean are mutually consistent, and find $\max_e |W_e - S_e Q_e| \leq 5.6 \times 10^{-17}$ across all capture cells. The identity is not expected to survive averaging across prompts: where a per-domain or overall W , S , and Q is reported (in the accompanying tables, not in this paper), each is a separate mean over the domain’s prompts and the model’s layers, so the tabulated S and Q are marginal means whose product differs from the tabulated W by the across-prompt S – Q covariance. Domain winners are ranked by W , the mean routed weight per token, which is well defined regardless of the decomposition.

The two passes. For each prompt we partition its token positions into the prefill block (the prompt tokens) and the generation block (the tokens the model produced). All routing statistics are computed within one block or the other, never pooled across them. For each domain we then form the per-domain mean of W_e over that domain’s three prompts and call the highest- W expert the *domain winner*. This yields, per model and per pass, a list of 20 domain winners. The analysis in this paper is almost entirely about how those two lists differ.

Concentration of a winner list. To summarize a list of 20 winners we use three numbers. The *distinct count* is how many different experts appear. *Max wins* is the number of domains taken by the single most frequent winner. And the normalized entropy of the winner distribution, $H = -\sum_i p_i \log_2 p_i / \log_2 20$ where p_i is the fraction of domains won by expert i , runs from 0 (one expert wins everything) to 1 (every domain has its own winner). These are deterministic summaries of the winner lists; they introduce no new model evaluation.

5 Results

5.1 Over prefill, one generalist wins almost everything

In the 35B model, the prefill domain-winner list is dominated by a single expert that takes 18 of the 20 domains. Only two domains escape it: chemistry and linguistics each have their own prefill winner. The same expert is also the top expert overall on the prefill block. By the summaries of Section 4, the prefill winner list has just 3 distinct experts and a normalized entropy of 0.13

Model	Pass	Distinct winners	Max wins	Herfindahl	Norm. entropy
Qwen 35B	prefill	3	18	0.815	0.13
Qwen 35B	generation	20	1	0.050	1.00
Qwen 122B	prefill	7	13	0.445	0.42
Qwen 122B	generation	18	2	0.060	0.95

Table 1: Concentration of the 20-domain winner list, by model and pass. Distinct winners and max wins are counts out of 20 domains. Herfindahl is $\sum_i p_i^2$ and normalized entropy is H from Section 4, both on the distribution of domains won per expert. Each row summarizes one domain-winner table from the source archive ($n = 60$ prompts per model, 3 per domain).

(Table 1). Read literally, the prefill routing assigns one expert to nearly every subject: a single high-frequency default standing in for the rest.

5.2 Over generation, the winners disperse

The generation domain-winner list for the same model and the same prompts diverges sharply from its prefill list. All 20 domains are won by 20 distinct experts; no expert wins more than one domain. The normalized entropy of the winner list rises from 0.13 to 1.00. The generalist that won 18 domains during prefill wins exactly one during generation (political science) and otherwise drops out of the per-domain lead. In its place, domain-specific experts surface. In philosophy, for instance, the generation winner held only rank 5 within philosophy during prefill ($W = 0.0188$ generation versus a mid-pack prefill weight), and it wins no other domain.

The dispersion holds regardless of where we cut the generated text. These models do not emit a clean end-of-turn token; they spell out the end-of-turn marker as literal text and then continue into hallucinated follow-up turns until the token cap. Trimming the generation block at that marker, or leaving it untrimmed, produces the same 20 distinct domain winners. The structure sits in the answer content.

5.3 Quantifying the shift

Table 1 collects the concentration summaries for both passes and both models, and Figure 1 shows the two leading numbers visually. The pattern is the same in both models and large in both: a concentrated, low-entropy winner list over prefill becomes a dispersed, high-entropy one over generation. The generalist’s reach (max wins) falls from 18 to 1 in the 35B model and from 13 to 2 in the 122B model.

Figure 2 restates the same shift on a single scale-free axis: the normalized entropy of the winner distribution moves from 0.13 to 1.00 in the 35B model and from 0.42 to 0.95 in the 122B model. The prefill values differ between the two models (the smaller model’s prefill routing is more concentrated than the larger model’s), but both land near the dispersed ceiling over generation.

5.4 Token count does not explain the shift

A first objection is that prefill and generation differ in length: a prompt is tens of tokens, an answer is hundreds, and longer token sets could disperse the top experts for purely statistical reasons. We address this with a token-balanced design. Three prompts are each packed with twenty domain

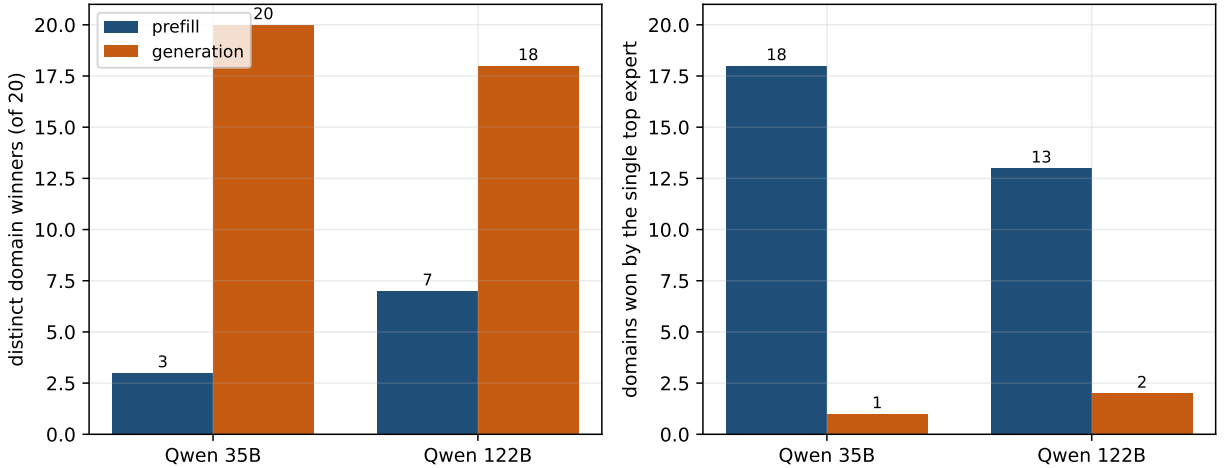


Figure 1: Left: number of distinct experts among the 20 domain winners. Right: number of domains won by the single most frequent winner. In both models the prefill winners collapse onto a generalist and the generation winners spread across the domains. $n = 60$ prompts per model (20 domains $\times$ 3).

questions and padded to exactly 446 prompt tokens, and each is allowed 2,048 generated tokens, so the prefill and generation blocks are matched in scale and the per-chunk comparisons use identical token budgets.

Within this design the prefill and generation *expert sets* barely overlap. Figure 3 reports the Jaccard overlap between the top experts of the two passes for each packed chunk: by routed weight it is 0.00, 0.23, and 0.07 across the three chunks, and by selection rate 0.07, 0.23, and 0.00. With token counts held fixed, the set of experts that dominates generation is mostly disjoint from the set that dominates prefill. The routing entropy itself is almost unchanged between the two passes (0.958 prefill, 0.953 generation): generation spreads each token over a similar number of experts and routes them to *different* ones. The regime moves the winners; length holds steady.

5.5 The pattern reproduces in a second model with different experts

The 122B model gives an independent replication. Its expert indices are unrelated to the 35B model’s, and indeed a different generalist emerges: over prefill, one expert wins 13 of 20 domains (with a second taking 2 more), for 7 distinct winners and entropy 0.42, less extreme than the 35B model and still strongly concentrated. Over generation the winners disperse to 18 distinct experts across the 20 domains, entropy 0.95 (Table 1, Figure 1). The global top experts also change identity between passes: the prefill leaders are one set of indices and the generation leaders are a different set. The specific experts change from model to model; the prefill-concentrated, generation-dispersed shape carries over.

The two models also illustrate that an expert’s role can move between passes within a single model. In the 122B model the expert that leads philosophy *generation* is mid-pack during philosophy prefill, while the expert that happened to look philosophy-relevant in the smaller model’s generation has, in the larger model, its only strong footprint in a different domain (computer science). We read these as cautions against naming experts from any single pass or carrying an index across models.

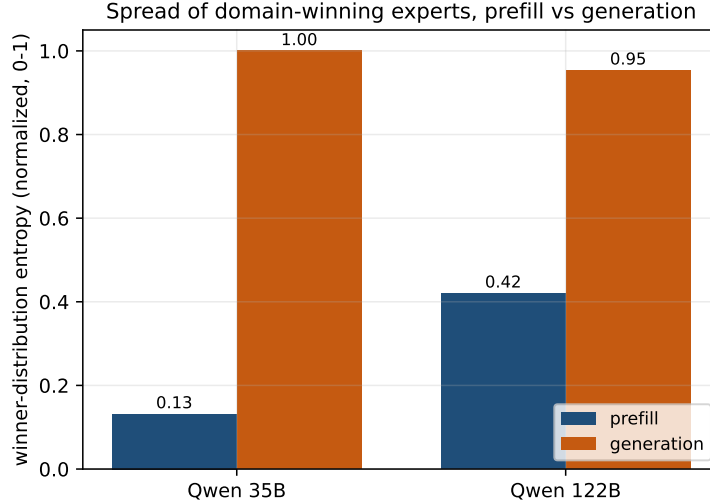


Figure 2: Normalized entropy of the 20-domain winner distribution (0 = one expert wins all domains, 1 = every domain has its own winner), by model and pass. Both models move from a concentrated prefill winner list to a dispersed generation one.

6 Interpretation

The measurement supports a narrow, mechanical claim. In these models the routing over prompt positions is dominated by a few high-frequency experts and carries little domain signal, while the routing over generated positions resolves into distinct per-domain leaders. Domain specialization, to the extent the router expresses it at all, is largely a property of the generation pass.

A plausible reading of the prefill behavior gives the field’s negative results a principled cause. Reading a short prompt is a fairly generic operation (tokenizing, parsing, building context) for which the router falls back on common, broadly useful experts; domain-specific computation is recruited once the model commits to producing domain-specific content token by token. On that reading, supplied-text routing is bound to miss domain structure, because it observes the model only during the pass that has yet to recruit the domain machinery. This is a hypothesis about the cause. The data locate the effect; its cause stays open.

A deflationary reading would credit the dispersion to the generation pass in general: perhaps generation simply routes more diffusely, and the spread of winners follows from that. Two entropies, measured together in the token-balanced control, tell against it. The per-token routing entropy holds nearly steady across passes (0.958 prefill, 0.953 generation): the model spreads each token’s weight over a similar number of experts in both regimes. The winner-distribution entropy moves far, from 0.13 to 1.00 in the 35B model and from 0.42 to 0.95 in the 122B model. The per-token spread stays put while the identity of each domain’s leading expert changes. The generation pass relabels the domain leaders, and that relabeling is the specialization signal.

The implication corrects the reading the field has drawn. An analysis that characterizes expert–topic structure from prefill routing, or from routing pooled over a sequence without separating the passes, sees the concentrated prefill picture and understates specialization. The same probe, read over the generation pass, recovers per-domain structure. So a negative result obtained over supplied text reports the prefill half accurately and leaves the generation half unmeasured; when the goal is to ask what an MoE’s experts specialize in, the generation pass is the place to measure, and the

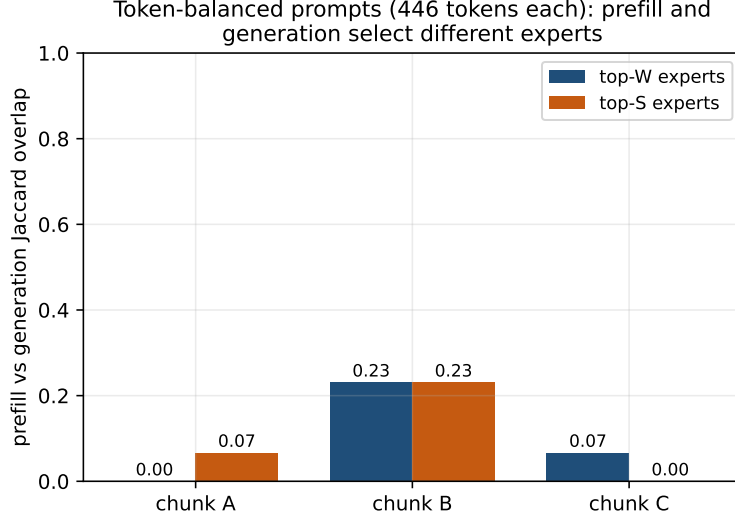


Figure 3: Token-balanced control (35B model, three packed prompts of 446 tokens each). Jaccard overlap between the prefill and generation top-expert sets, by routed weight and by selection rate. With token counts matched, prefill and generation select largely disjoint experts.

two passes should be reported separately.

7 Limitations and falsification tests

What the present evidence does not yet settle.

- **Cross-family scope is the boundary of the sharp claim.** Both checkpoints are 8-bit quantized HauhouCS fine-tunes of Qwen3.5 MoE models. The pattern reproduces across two sizes with non-corresponding indices, which carries the model-independent, method-level reading: supplied-text routing understates specialization. Turning that reading into a statement about any one prior result requires re-measuring that result’s setup. Concretely, re-running the Mixtral routing analysis of Jiang et al. (2024) over the generation pass would upgrade the corrective from inference to demonstration; we predict it would resolve the topic structure their supplied-text analysis did not show. Until such a test, the demonstrated claim covers this lineage and the correction applies to the measurement convention.
- **Generation length and the token cap.** Generated blocks are long and frequently reach the cap (in the 122B run, 49 of 60 cells hit 2,048 tokens; mean generated length $\approx 1,868$), and every cell eventually spills into chat-template continuation. Trimming at the spill boundary leaves the 35B winner list unchanged, which argues against a pure spill artifact, but a length- and cap-matched generation budget per prompt would make the comparison cleaner.
- **“Domain winner” is a coarse statistic.** It reduces a domain to its single top expert by mean weight. A domain with two near-tied leading experts and a domain with one dominant expert are treated the same. Reporting the full top- k per domain, and the stability of the winner across the three prompts, would give a fuller picture than the winner alone.

- **Significance is asserted descriptively.** The concentration summaries describe the observed winner lists; we have not attached a randomization test that asks how often a winner list this dispersed would arise under a null of pass-independent routing. The 18-versus-20 and 13-versus-18 contrasts are large, but a permutation test would quantify them.
- **Three prompts per domain.** Twenty domains at three prompts each is enough to expose the prefill/generation contrast but thin for per-domain claims. The per-domain winner is the least reliable level of this analysis.

Tests that would confirm or break the claim.

- Re-run the Mixtral analysis of Jiang et al. (2024) — the same data and the same routing readout — over generated tokens as well as supplied text. The corrective predicts that the topic structure absent over supplied text resolves over generation; this is the direct test of the correction and the experiment that would carry it from a method-level inference to a model-specific demonstration.
- Run the same probe on a base (non-fine-tuned) Qwen3.5 MoE and on an MoE from a different family (for example a different vendor’s top- k router); the claim predicts the same prefill-concentrated, generation-dispersed shift with, again, non-transferable indices.
- Fix a common generation budget per prompt and re-measure, removing the cap and the variable-length confound entirely.
- Replace the single-winner statistic with a permutation test that shuffles the prefill/generation labels on token blocks and asks how often the dispersion gap seen here arises by chance; a corrective aimed at the field’s negative results invites this stricter standard of evidence.
- Increase prompts per domain (say 10–20) and report winner stability, so per-domain assignments carry a confidence; the higher-stakes framing makes that per-domain reliability worth the additional captures.
- Track, position by position within the generation block, when a domain’s generation winner overtakes the prefill generalist; this would test the “specialization is recruited as content is produced” reading directly.

8 Conclusion

Two MoE language models route prompt tokens and generated tokens to different experts, and the difference is structured. Over the prompt-reading pass a single generic expert wins almost every domain; over the generation pass the domains acquire distinct leading experts. The shift is large, survives matching prompts to a common length, and reproduces across two model sizes whose expert indices do not correspond. The lesson is a correction. The widely cited negative result on MoE expert specialization was measured over supplied text, the prefill half of the forward pass, the obvious and cheap place to look, and the half where the structure stays hidden. Prefill is the half the field measured; generation is the half it missed, and the half where specialization resolves. For anyone asking what an MoE’s experts specialize in, separate the two passes and read specialization from generation.

Reproducibility and audit note

The numbers in this paper are transcribed from per-domain winner tables in an existing capture archive and are listed claim-by-claim, with source files, in the accompanying `SOURCES.md`. The three figures and the concentration summaries in Table 1 are produced by `make_figures.py` from the winner lists alone; that script re-runs no model and performs only the deterministic counting described in Section 3. The underlying captures were produced with greedy decoding and a top-8-of-256 routing reconstruction whose $W = S \cdot Q$ identity holds per token set at machine precision (residual $\leq 5.6 \times 10^{-17}$); we use the reconstructed, log-derived routing tables rather than any earlier hand-summarized figures. Two points of provenance are worth stating plainly: the second model’s analysis was framed in its source notes around locating an analog of one particular expert, and we have deliberately set that framing aside and report only the pass-dependent concentration that the same tables support; and the per-domain winner is, as Section 7 notes, the least reliable layer of the analysis, so the claims we rest on are the aggregate concentration contrasts, not any single domain’s assignment.

Data and Code Availability

The data supporting this paper accompany it in the `data/` directory, organized as three runs: the 35B 60-prompt primary probe, the 35B token-balanced 3-chunk control, and the 122B replication (prefill and generation captures). Each run folder contains the per-domain winner tables from which every number in the paper is taken; a manifest (`data/README.md`) maps each run to the claims it supports. The three figures and the concentration summaries in Table 1 are reproduced from those winner lists by `make_figures.py`, which re-runs no model. A claim-by-claim source index is given in `SOURCES.md`. The raw router tensors (`.npz`) are included as non-load-bearing provenance. No new model training was performed.

AI Collaboration Statement

Generative AI (Anthropic’s Claude) was used as a tool in preparing this manuscript: to assist with drafting and revising prose, organizing the manuscript, and locating and formatting the bibliography. All experiments, captures, and numerical results are the author’s; every reported value was traced to a source capture (see `SOURCES.md`), and every reference was verified by the author against its authoritative record to exist and to support the statement it is attached to. No AI system is an author. The author reviewed the entire text and takes full responsibility for its content, including any errors.

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This work received no external funding. The author declares no competing interests. The models studied are third-party fine-tunes used as-is for measurement; the author has no affiliation with their creators.

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